

Week-06 HandsOn Solution

❖ ReactJS-HOL

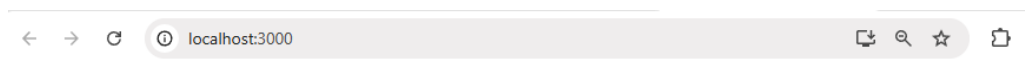
Question 1: Introduction to React & Creating First App

- Code:

```
import logo from './logo.svg';
import './App.css';

function App() {
  return (
    <h1>Welcome the First Session of React</h1>
  );
}
export default App;
```

- Output:



Welcome the First Session of React

Question 2: Class Components in React

- Code:

//About.js

```
import React, { Component } from 'react';
class About extends Component {
  render() {
    return (
      <div>
        <h3>Welcome to the About page of Student Mangement Portal </h3>
      </div>
    );
  }
}
export default About;
```

//Contact.js

```
import React, { Component } from 'react';
class Contact extends Component {
  render() {
    return (
      <div>
        <h3>Welcome to Contact page of Student Mangement Portal</h3>
      </div>
    );
  }
}
export default Contact;
```

//Home.js

```
import React, { Component } from 'react';
class Home extends Component {
  render() {
```

```

    return (
      <div>
        <h3>Welcome to the Home Page of Student Management Portal </h3>
      </div>
    );
  }
}
export default Home;

```

• Output:



Welcome to the Home Page of Student Management Portal

Welcome to the About page of Student Mangement Portal

Welcome to Contact page of Student Mangement Portal

Question 3: Functional Components & Styling

• Code:

```

import './Stylesheets/mystyle.css'

const percentToDecimal = (decimal) => {
  return (decimal.toFixed(2) + '%')
}

const calcScore = (total, goal) => {
  return percentToDecimal(total / goal)
}

export const CalculateScore = ({ Name, School, total, goal }) => (
  <div className='formalstyle'>
    <h1><font color="Brown">Student Details : </font></h1>
    <div className='Name'>
      <b> <span>Name : </span></b>
      <span>{Name}</span>
    </div>
    <div className='School'>
      <b><span>School : </span></b>
      <span>{School}</span>
    </div>
    <div className='Total'>
      <b><span>Total : </span></b>
      <span>{total}</span>
      <span>Marks</span>
    </div>
    <div className='Score'>
      <b>Score : </b>
      <span>{calcScore(total, goal)}</span>
    </div>
  </div>)

```

• Output:



Student Details :

Name : Steeve

School : DNV Public School

Total : 284Marks

Score : 94.67%

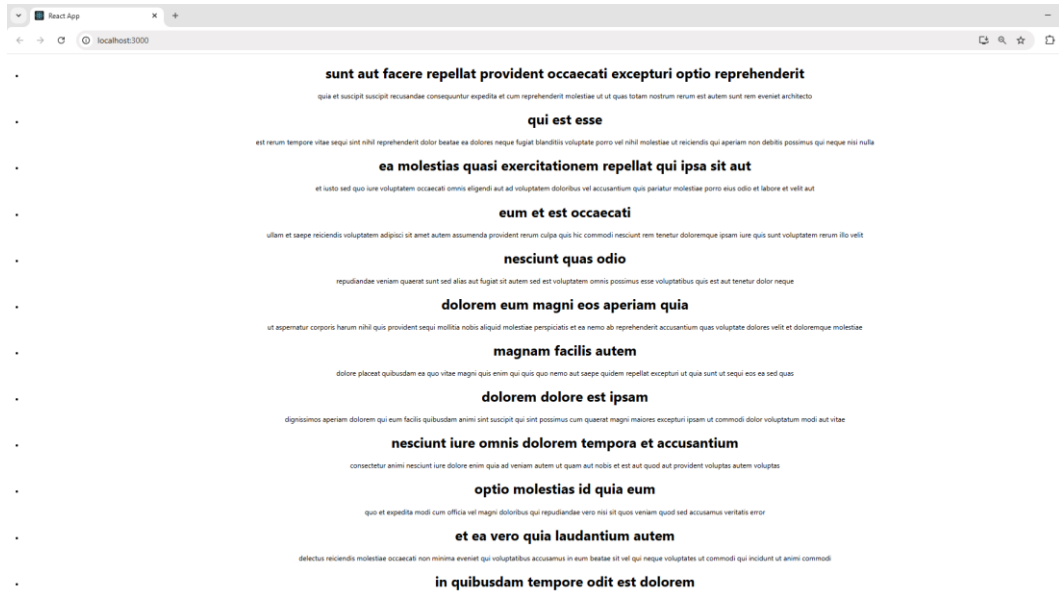
Question 4: React Component Lifecycle Methods

- Code:

```
import Post from './Post';
import React from 'react';
class Posts extends React.Component {
  constructor(props) {
    super(props);
    this.state = {
      posts: []
    };
  }
  loadPosts() {
    fetch('https://jsonplaceholder.typicode.com/posts')
      .then((response) => response.json())
      .then((data) => {
        const obj = data.map(value => new Post(value.id, value.title, value.body));
        this.setState({ posts: obj });
      })
      .catch((error) => {
        console.error("Error in Fetching : ", error);
      });
  }
  componentDidMount() {
    this.loadPosts();
  }
  render() {
    const { posts } = this.state;

    return (
      <>
        <ul>
          {posts.map(post => (
            <li key={post.id}>
              <h1>{post.title}</h1>
              <p>{post.body}</p>
            </li>
          ))}
        </ul>
      </>
    );
  }
  componentDidCatch(error, info) {
    console.error("Error in Posts Component: ", error);
  }
}
export default Posts;
```

- Output:



Question 5: Styling React Components with CSS Modules

• Code:

//App.js

```
import logo from './logo.svg';
import './App.css';
import { CohortsData } from './Cohort'
import CohortDetails from './CohortDetails';

function App() {

  return (
    <div>
      <h1>Cohorts Details</h1>
      {CohortsData.map(cohort => <CohortDetails cohort={cohort}/>)}
    </div>
  );
}

export default App;
```

//Cohort.js

```
class Cohort {
  constructor(cohortCode,
    startDate,
    technology,
    trainerName,
    coachName,
    currentStatus) {
    this.cohortCode = cohortCode;
    this.coachName = coachName;
    this.trainerName = trainerName;
    this.technology = technology;
    this.startDate = startDate;
    this.currentStatus = currentStatus;
  }
}

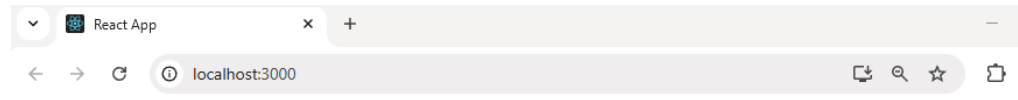
const CohortsData = [
  new Cohort('INTADMDF10', '22-Feb-2022', '.NET FSD', 'Jojo Jose', 'Aathma', 'Scheduled'),
  new Cohort('ADM21JF014', '10-Sep-2021', 'Java FSD', 'Elisa Smith', 'Apoorv', 'Ongoing'),
  new Cohort('CDBJF21025', '24-Dec-2021', 'Java FSD', 'John Doe', 'Aathma', 'Ongoing'),
  new Cohort('INTADMJF12', '22-Feb-2022', 'Java FSD', 'To Be Assigned', 'Ibrahim', 'Scheduled'),
```

```

new Cohort('CDE22JF011', '24-Dec-2021', 'Java FSD', 'Emma Swan', 'Apoorv', 'Ongoing'),
new Cohort('INTADMDF09', '22-Feb-2022', 'Dataware Housing', 'Babjee Rao', 'Aathma', 'Scheduled'),
new Cohort('ADM22DF001', '10-Sep-2021', '.NET FSD', 'Marie Curie', 'Ibrahim', 'Ongoing'),
];
export { Cohort, CohortsData };

```

• Output:



Cohorts Details

INTADMDF10 -.NET FSD

Started On
22-Feb-2022
Current Status
Scheduled
Coach
Aathma
Trainer
Jojo Jose

ADM21JF014 -Java FSD

Started On
10-Sep-2021
Current Status
Ongoing
Coach
Apoorv
Trainer
Elisa Smith

CDBJF21025 -Java FSD

Started On
24-Dec-2021
Current Status
Ongoing
Coach
Aathma
Trainer
John Doe

INTADMJF12 -Java FSD

Started On
22-Feb-2022
Current Status
Scheduled
Coach
Ibrahim
Trainer
To Be Assigned

CDE22JF011 -Java FSD

Started On
24-Dec-2021
Current Status
Ongoing
Coach
Apoorv
Trainer
Emma Swan

INTADMDF09 -Dataware Housing

Started On
22-Feb-2022
Current Status
Scheduled
Coach
Aathma
Trainer
Babjee Rao

ADM22DF001 -.NET FSD

Started On
10-Sep-2021
Current Status
Ongoing
Coach
Ibrahim
Trainer
Marie Curie

Week-07 HandsOn Solution

❖ ReactJS-HOL

Question 9: ES6 Features with Map, Arrow Functions, and Destructuring in React Cricket App.

• Code:

```
import React, { useState } from 'react';
import './App.css';

function App() {
  const [flag, setFlag] = useState(true);

  return (
    <div className="App">
      <h1>Cricket Players Dashboard</h1>
      <button onClick={() => setFlag(!flag)}>
        Toggle View ({flag ? "List View" : "Indian Players"})
      </button>

      {flag ? <ListofPlayers /> : <IndianPlayers />}
    </div>
  );
}

function ListofPlayers() {
  const players = [
    { id: 1, name: "Virat Kohli", score: 120 },
    { id: 2, name: "Rohit Sharma", score: 65 },
    { id: 3, name: "KL Rahul", score: 45 },
    { id: 4, name: "Rishabh Pant", score: 78 },
    { id: 5, name: "Hardik Pandya", score: 85 },
    { id: 6, name: "Jasprit Bumrah", score: 12 },
    { id: 7, name: "Ravindra Jadeja", score: 55 },
    { id: 8, name: "Mohammed Shami", score: 8 },
    { id: 9, name: "Shreyas Iyer", score: 92 },
    { id: 10, name: "Yuzvendra Chahal", score: 15 },
    { id: 11, name: "Bhuvneshwar Kumar", score: 25 }
  ];

  const lowScorers = players.filter(player => player.score < 70);

  return (
    <div>
      <h2>All Players</h2>
      <ul className="player-list">
        {players.map(player => (
          <li key={player.id}>
            {player.name} - <span className="score">{player.score}</span>
          </li>
        ))}
      </ul>

      <h2>Players with Score Below 70</h2>
      <ul className="player-list">
        {lowScorers.map(player => (
          <li key={player.id}>
            {player.name} - <span className="low-score">{player.score}</span>
          </li>
        ))}
      </ul>
    </div>
  );
}
```

```

}

function IndianPlayers() {
  const players = ["Virat", "Rohit", "Dhoni", "Rahul", "Pant", "Bumrah", "Jadeja"];

  const [first, second, ...others] = players;
  const oddPlayers = players.filter((_, index) => index % 2 === 0);
  const evenPlayers = players.filter((_, index) => index % 2 !== 0);

  const t20Players = ["Hardik", "Chahal", "Surya"];
  const ranjiPlayers = ["Pujara", "Rahane", "Vihari"];
  const allPlayers = [...t20Players, ...ranjiPlayers];

  return (
    <div>
      <h2>Indian Players Analysis</h2>

      <h3>Destructured Players</h3>
      <p>First: {first}, Second: {second}, Others: {others.join(", ")}</p>

      <h3>Odd vs Even Team</h3>
      <div className="team-container">
        <div>
          <h4>Odd Team</h4>
          <ul>
            {oddPlayers.map((player, index) => (
              <li key={index}>{player}</li>
            ))}
          </ul>
        </div>
        <div>
          <h4>Even Team</h4>
          <ul>
            {evenPlayers.map((player, index) => (
              <li key={index}>{player}</li>
            ))}
          </ul>
        </div>
      </div>

      <h3>Merged Players (T20 + Ranji)</h3>
      <ul>
        {allPlayers.map((player, index) => (
          <li key={index}>{player}</li>
        ))}
      </ul>
    </div>
  );
}

export default App;

```

• Output:



Question 10: JSX & Inline CSS in React

- **Code:**

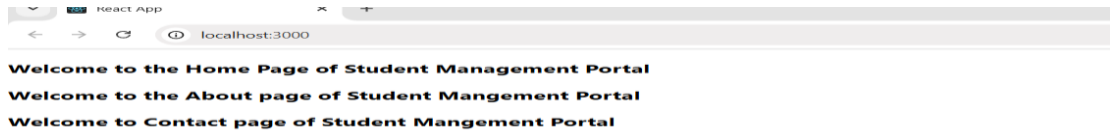
```
import React from 'react';
import './App.css';

function App() {
  const offices = [
    { id: 1, name: "Downtown Office", rent: 55000, address: "123 Main St", image:
"https://via.placeholder.com/300x200?text=Office+1" },
    { id: 2, name: "Tech Park Suite", rent: 75000, address: "456 Tech Ave", image:
"https://via.placeholder.com/300x200?text=Office+2" },
    { id: 3, name: "Executive Plaza", rent: 65000, address: "789 Business Blvd", image:
"https://via.placeholder.com/300x200?text=Office+3" },
    { id: 4, name: "Innovation Hub", rent: 45000, address: "321 Creative Lane", image:
"https://via.placeholder.com/300x200?text=Office+4" }
  ];

  return (
    <div className= "App" >
      <h1>Office Space Rental </h1>
      <div className = "office-list" >
        {
          offices.map(office => (
            <div key= { office.id } className = "office-card" >
              <img src={ office.image } alt = { office.name } />
              <h2>{ office.name } </h2>
              <p> Address: { office.address } </p>
              <p style = {{ color: office.rent < 60000 ? 'red' : 'green', fontWeight: 'bold' }} >
                Rent: ₹{ office.rent.toLocaleString() }
              </p>
            </div>
          ))
        }
      </div>
    </div>
  );
}

export default App;
```

- **Output:**



Question 11: Event Handling in React

- Code:

```
import React, { useState } from 'react';
import './App.css';

function App() {
  const [count, setCount] = useState(0);
  const [message, setMessage] = useState("");
  const [rupees, setRupees] = useState("");
  const [euros, setEuros] = useState("");

  const increment = () => {
    setCount(count + 1);
    sayHello();
  };

  const sayHello = () => {
    setMessage('Hello! Welcome to React Events!');
  };

  const sayWelcome = (msg) => {
    setMessage(msg);
  };

  const handleClick = (e) => {
    console.log(e);
    setMessage('I was clicked!');
  };

  const handleConversion = () => {
    const rate = 0.011; // 1 INR = 0.011 EUR
    setEuros((rupees * rate).toFixed(2));
  };

  return (
    <div className= "App" >
      <h1>React Event Examples </h1>

      < div className = "counter-section" >
        <h2>Counter: { count } </h2>
        < button onClick = { increment } > Increment(+1 and Say Hello) </button>
        < button onClick = { () => setCount(count - 1) } > Decrement </button>
      </div>

      < div className = "message-section" >
        <h3>{ message } </h3>
        < button onClick = { () => sayWelcome("Welcome to React!") } > Say Welcome </button>
        < button onClick = { handleClick } > Click Me(Synthetic Event) </button>
      </div>

      < div className = "converter-section" >
        <h2>Currency Converter </h2>
        < div >
          <label>
```

```

      Indian Rupees:
      <input
        type="number"
        value = { rupees }
        onChange = {(e) => setRupees(e.target.value)}
      />
    </label>
    < button onClick = { handleConversion } > Convert to Euros </button>
  </div>
  { euros && <p>₹ { rupees } = € { euros } </p> }
</div>
</div>
);
}

export default App;

```

• Output:



Student Details :

Name : Steeve

School : DNV Public School

Total : 284Marks

Score : 94.67%

Question 12: Conditional Rendering with Login/Logout in React Ticket Booking App.

• Code:

```

import React, { useState } from 'react';
import './App.css';

function App() {
  const [isLoggedIn, setIsLoggedIn] = useState(false);
  const flights = [
    { id: 1, flightNo: 'AI101', from: 'Delhi', to: 'Mumbai', time: '10:00 AM' },
    { id: 2, flightNo: 'AI202', from: 'Mumbai', to: 'Chennai', time: '12:30 PM' },
    { id: 3, flightNo: 'AI303', from: 'Bangalore', to: 'Delhi', time: '03:15 PM' }
  ];

  return (
    <div className= "App" >
      <h1>Flight Ticket Booking </h1>

      < button
        onClick = {( ) => setIsLoggedIn(!isLoggedIn)}
      >
        { isLoggedIn? 'Logout': 'Login' }
      </button>

      {
        isLoggedIn ? (
          <div className= "user-view" >
            <h2>Welcome User! </h2>

```

```

    <div className = "booking-form" >
      <h3>Book Your Flight </h3>
      <select>
        {
          flights.map(flight => (
            <option key= { flight.id } value = { flight.flightNo } >
              { flight.flightNo } : { flight.from } to { flight.to } ({ flight.time })
            </option>
          ))
        }
      </select>
      <input type = "date" placeholder = "Travel Date" />
      <input type="number" placeholder = "Passengers" min = "1" />
      <button className="book-btn" > Book Now </button>
    </div>
  </div>
) : (
  <div className= "guest-view" >
    <h2>Available Flights </h2>
    <table >
      <thead>
        <tr>
          <th>Flight No </th>
          <th > From </th>
          <th > To </th>
          <th > Time </th>
        </tr>
      </thead>
      <tbody>
        {
          flights.map(flight => (
            <tr key= { flight.id } >
              <td>{ flight.flightNo } </td>
              <td > { flight.from } </td>
              <td > { flight.to } </td>
              <td > { flight.time } </td>
            </tr>
          ))
        }
      </tbody>
    </table>
    <p > Please login to book tickets </p>
  </div>
)
</div>
);
}

```

export default App;

• Output:



Question 13: Conditional Rendering in React

• Code:

```
import React, { useState } from 'react';
import './App.css';

function App() {
  const [activeTab, setActiveTab] = useState('books');

  // Sample data
  const books = [
    { id: 1, title: 'React Basics', author: 'John Doe', pages: 200 },
    { id: 2, title: 'Advanced JavaScript', author: 'Jane Smith', pages: 350 }
  ];

  const blogs = [
    { id: 1, title: 'Getting Started with React', author: 'Alex Johnson', date: '2023-05-15' },
    { id: 2, title: 'ES6 Features You Should Know', author: 'Sarah Williams', date: '2023-06-20' }
  ];

  const courses = [
    { id: 1, title: 'React Fundamentals', duration: '4 weeks', level: 'Beginner' },
    { id: 2, title: 'Advanced React Patterns', duration: '6 weeks', level: 'Advanced' }
  ];

  return (
    <div className= "App" >
      <h1>Blogger App </h1>

      <div className = "tabs" >
        <button
          onClick={ () => setActiveTab('books') }
          className= { activeTab === 'books' ? 'active' : "
        } >
        Books
      </button>
      <button
        onClick = { () => setActiveTab('blogs')}
        className = { activeTab === 'blogs' ? 'active' : "" }
      >
        Blogs
      </button>
      <button
        onClick = { () => setActiveTab('courses')}
        className = { activeTab === 'courses' ? 'active' : "" }
      >
        Courses
      </button>
    </div>

    <div className = "content" >
      { activeTab === 'books' && (
        <div className="book-details" >
          <h2>Book Details </h2>
          <ul>
            {
              books.map(book => (
                <li key= { book.id } >
                  <h3>{ book.title } </h3>
                  <p > Author: { book.author } </p>
                  <p > Pages: { book.pages } </p>
                </li>
              ))
            }
          </ul>
        </div>
      )}

      {
        activeTab === 'blogs' && (
          <div className="blog-details" >
            <h2>Blog Details </h2>
            <ul>
              {
                blogs.map(blog => (
```

```

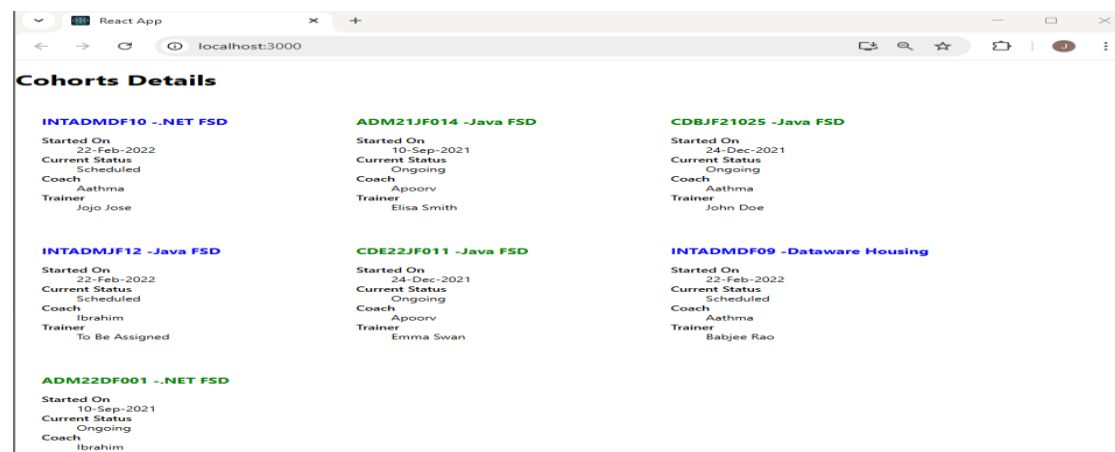
    <li key= { blog.id } >
      <h3>{ blog.title } </h3>
      <p> By { blog.author } on { blog.date } </p>
    </li>
  ))
}
</ul>
</div>
)
}

{
  activeTab === 'courses' && (
    <div className="course-details" >
      <h2>Course Details </h2>
      <table>
        <thead>
          <tr>
            <th>Title </th>
            <th> Duration </th>
            <th> Level </th>
          </tr>
        </thead>
        <tbody>
          {
            courses.map(course => (
              <tr key= { course.id } >
                <td>{ course.title } </td>
                <td> { course.duration } </td>
                <td> { course.level } </td>
              </tr>
            ))
          }
        </tbody>
      </table>
    </div>
  )
}
</div>
</div>
);
}

export default App;

```

1. Output:



Cohorts Details		
INTADMDF10 -.NET FSD Started On 22-Feb-2022 Current Status Scheduled Coach Aathma Trainer Jojo Jose	ADM21JF014 -Java FSD Started On 10-Sep-2021 Current Status Ongoing Coach Apoorv Trainer Elisa Smith	CDBJF21025 -Java FSD Started On 24-Dec-2021 Current Status Ongoing Coach Aathma Trainer John Doe
INTADMJF12 -Java FSD Started On 22-Feb-2022 Current Status Scheduled Coach Ibrahim Trainer To Be Assigned	CDE22JF011 -Java FSD Started On 24-Dec-2021 Current Status Ongoing Coach Apoorv Trainer Emma Swan	INTADMDF09 -Dataware Housing Started On 22-Feb-2022 Current Status Scheduled Coach Aathma Trainer Babjee Rao
ADM22DF001 -.NET FSD Started On 10-Sep-2021 Current Status Ongoing Coach Ibrahim		